Recap: Differentiate: $f'(c) = \lim_{x \to c} \frac{f(x) - f(c)}{x - c}$

$$= \lim_{h \to 0} \frac{f(c+h) - f(c)}{h}$$

Theorem 3.1: Differentiable implies cts

If $f: D \to \mathbb{R}$ is differentiable at a point $c \in D$, then it is cts at $c \in D$.

Proof: Let $f$ be differentiable at $c \in D$, then

$$f'(c) = \lim_{x \to c} \left( \frac{f(x) - f(c)}{x - c} \right).$$

Hence, $\lim_{x \to c} \left( f(x) - f(c) \right) = \lim_{x \to c} \left( \frac{f(x) - f(c)}{x - c} (x - c) \right)$

$$= \lim_{x \to c} \left( \frac{f(x) - f(c)}{x - c} \right) \lim_{x \to c} (x - c)$$

$$= 0 \cdot f'(c)$$

$$= 0.$$

Hence, $f$ is cts at $c \in D$ since $\lim_{x \to c} f(x) = f(c)$. ☐

**Ex:** (Converse of Theorem 3.1)

Consider the absolute value function $|\cdot| : \mathbb{R} \to \mathbb{R}$ given by

$$|x| = \begin{cases} x : x \geq 0 \\ -x : x < 0 \end{cases}.$$

This is cts everywhere but not diff. at $c = 0$.

**Proof:** The right sided limit of the derivative at 0 is

$$\lim_{h \to 0^+} \left( \frac{|h| - |0|}{h} \right) = \lim_{h \to 0^+} \left( \frac{h}{h} \right) = 1.$$

The left sided limit gives

$$\lim_{h \to 0^-} \left( \frac{|h| - |0|}{h} \right) = \lim_{h \to 0^-} \left( \frac{-h}{h} \right) = -1.$$

Clearly the left sided limit doesn't equal the right sided limit and hence, the derivative doesn't exist at $c = 0$. ☐

## Theorem 3.3 : Algebra of differentiable functions

Let $f: D \to \mathbb{R}$ and $g: D \to \mathbb{R}$ both differentiable at $c \in D$. Then,

(i) $(f+g)'(c) = f'(c) + g'(c)$

(ii) $(\lambda f)'(c) = \lambda f'(c), \quad \forall \lambda \in \mathbb{R}$.

Proof : i) Since $f$ and $g$ are diff. at $c \in D$ it follows by the algebra of limits that

$$\lim_{x \to c} \left( \frac{f(x)+g(x) - f(c) - g(c)}{x - c} \right) = \lim_{x \to c} \left( \frac{f(x)-f(c)}{x-c} \right) + \lim_{x \to c} \left( \frac{g(x)-g(c)}{x-c} \right)$$

$$= f'(c) + g'(c). \quad \square$$

ii) Exercise.

## Theorem 3.4 : (Product Rule)

Let $f: D \to \mathbb{R}$ and $g: D \to \mathbb{R}$ both be differentiable

at $c \in D$. Then,

$$(fg)'(c) = f'(c)g(c) + g'(c)f(c).$$

Proof: Clearly,

$$\frac{f(x)g(x) - f(c)g(c)}{x-c} = \frac{f(x)g(x) + f(x)g(c) - f(x)g(c) - f(c)g(c)}{x-c}$$

Hence,

$$\lim_{x\to c}\left(\frac{f(x)g(x) - f(c)g(c)}{x-c}\right) = \lim_{x\to c}\left(f(x)\left(\frac{g(x)-g(c)}{x-c}\right) + g(c)\left(\frac{f(x)-f(c)}{x-c}\right)\right)$$

$$= \lim_{x\to c}(f(x))\, g'(c) + g(c)f'(c)$$

$$= f(c)g'(c) + g(c)f'(c) \quad \square.$$

Quotient rule → See notes.
Chain Rule → See notes

} Proof non-examinable.

# Interior and Mean Value Theorem

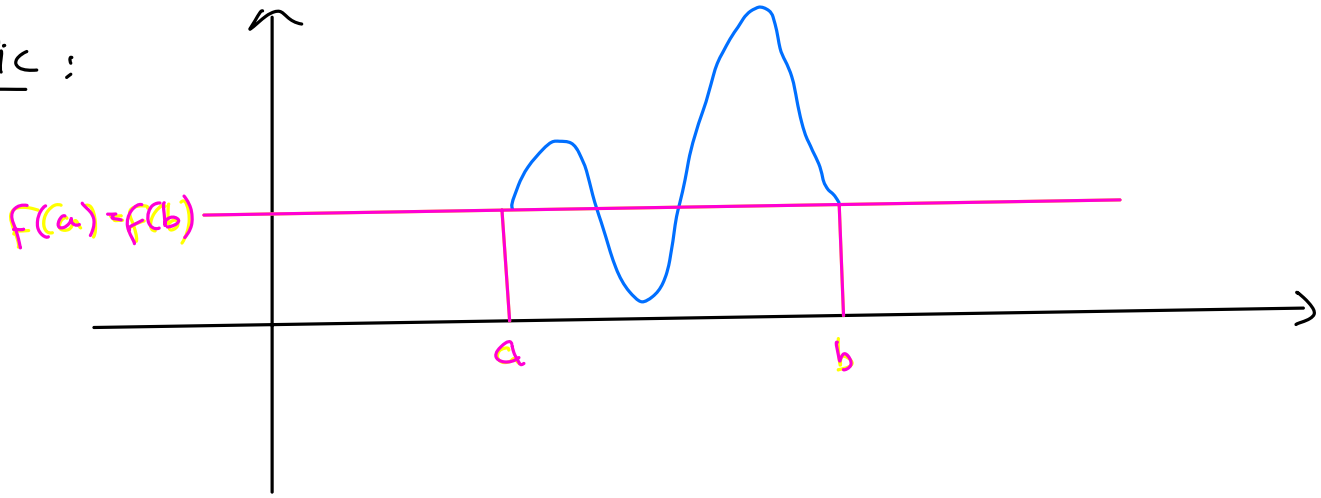
## Theorem 4.1 : Fermat's Theorem

Let $f: D \to \mathbb{R}$ be differentiable on some open interval $(a,b) \subset D$. Then, if $f$ attains a maximum or minimum value at some point $c \in (a,b)$, then $f'(c) = 0$.

Proof: Non-examinable

## Theorem 4.2 : Rolle's Theorem

Pic:




Let $f: [a,b] \to \mathbb{R}$ be cts on $[a,b]$ and diff. $(a,b)$.

Then, if $f(a) = f(b)$ there exists a $c \in (a,b)$ s.t $f'(c) = 0$.

Proof: Non-examinable

Example 4.3 → See notes.

## Theorem 4.4 : Mean Value Theorem

Let $f: [a,b] \to \mathbb{R}$. If $f$ is cts $[a,b]$ and diff. on $(a,b)$, there exists $c \in (a,b)$ s.t

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Pic :

